

Intelligent Systems for Industry, Supply Chain and Environment (in short “IS4”)

LESSON 1

Introduction and course description



Lesson outline

- Introduction
 - What is an intelligent system?
- Major AI Advancements, focus on the last 2 Years
- Organization of the course
- The exam format
 - Strategic approaches to the exam
- Biases
- Results and limits
- Applications of Intelligent systems for
 - industry
 - environment
 - business and medical applications
- **Main points**



Introduction to Intelligent systems

What is an intelligent system?

What can I do with them?



Glossary and acronyms

- AI: Artificial intelligence
- IS: Intelligent Systems
- ML: Machine Learning
- NN: Neural Network
- Train/Test: Training and testing (or Validation)
- HW/SW: Hardware/Software
- I/O: input/Output

What is an Intelligent System (IS)?

We can debate endlessly about a definition....

A (good) working description can be the following:

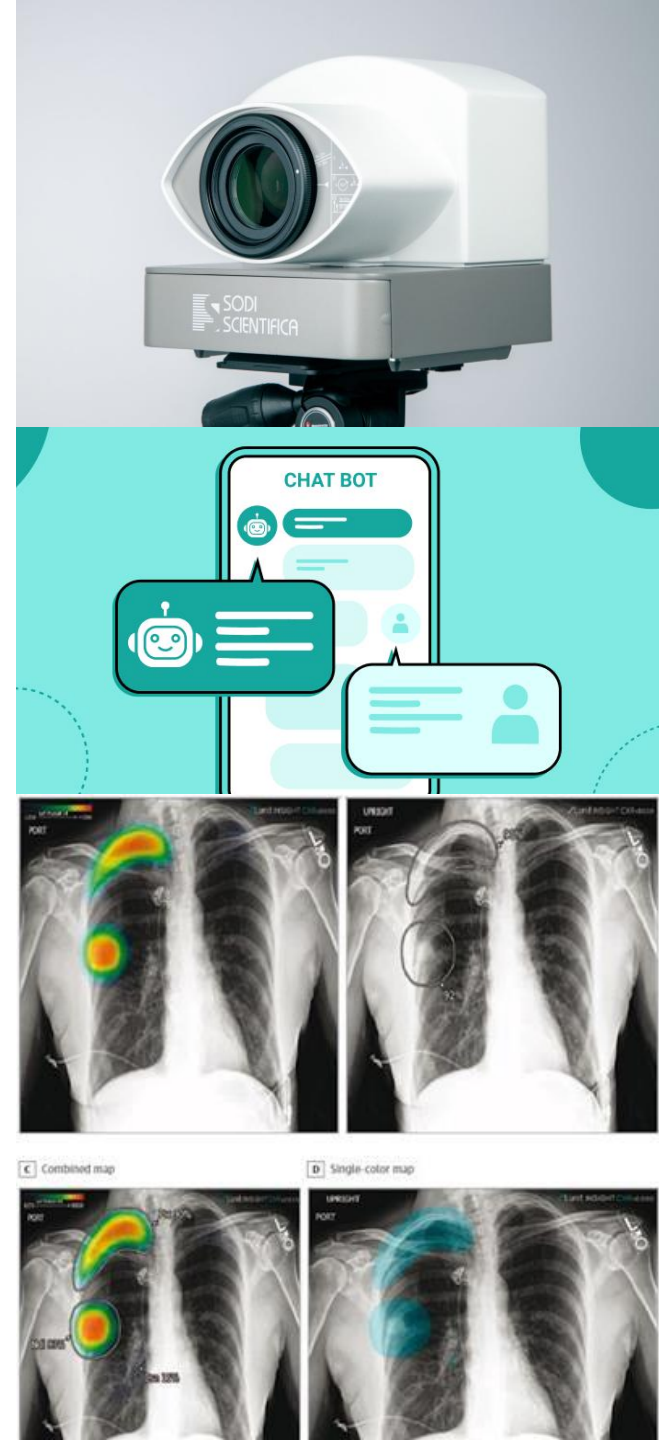
a computer-based system that aims to replicate human cognitive abilities such as

- *learning,*
- *perception,*
- *reasoning,*
- *and decision-making.*

By utilizing Machine Learning (ML), and other related technologies, these systems are capable of processing and analyzing data to perform tasks that typically require human intelligence, make predictions, or provide insights

Examples of Intelligent Systems

- Virtual assistants like Siri and Alexa
- Autonomous vehicles
- Medical diagnosis systems
- Recommendation systems
- Image recognition systems
- Chatbots
- Predictive maintenance systems
- Speech recognition systems
- Fraud detection systems
- Sentiment analysis systems
- And many many others...



Diffusion and relevance

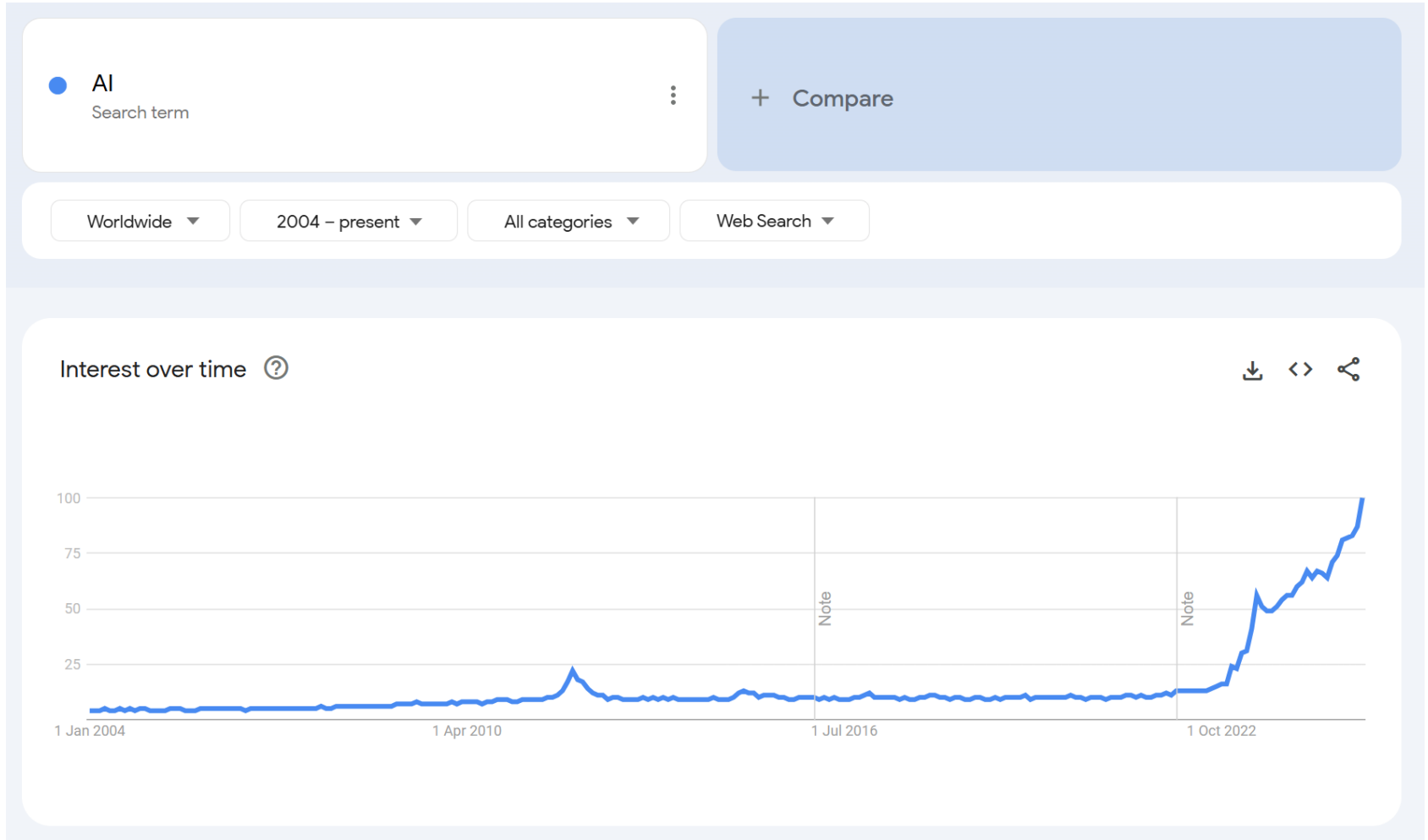
- Applications that sounded like **science fiction** only a decade ago are now part of our daily lives



- On a far larger scale, in the coming years AI will radically transform the future of work, the way in which economies function
- Machine learning in particular allows us to process data at unprecedented scales:
 - see patterns
 - detect problems earlier
 - and allocate resources more efficiently

AI trend

<https://trends.google.com/trends/explore?date=all&q=AI&hl=en-GB>



Sectors where AI is applied (a short list)



- Aviation
- Computer science
- Education
- Finance:
 - Algorithmic trading,
 - Market analysis and data mining,
 - Personal finance,
 - Portfolio management,
 - Credit underwriting
- Heavy industry
- Hospitals and medicine
- Marketing
- Media
- Music
- News, publishing and writing
- Online and telephone customer service
- Telecommunications maintenance
- Toys and games
- Transportation

Examples of **task** solved with AI

- Optical character recognition
 - Handwriting recognition
 - Speech recognition
 - Face recognition
 - Artificial creativity
 - Computer vision, Virtual reality and Image processing
 - Diagnosis (artificial intelligence)
 - Game theory and Strategic planning
 - Game artificial intelligence and Computer game bot
 - Natural language processing, Translation and Chatterbots
 - Nonlinear control and Robotics
- 
- A close-up photograph of a person's face, specifically the lower half, looking down at a smartphone held in their hand. The person has dark, curly hair. The background is a plain, light-colored wall.

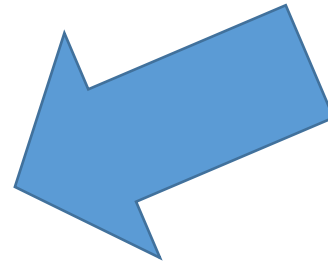


Applications of AI in Computer Science

- Web search, Computational biology, Finance, E-commerce, Space exploration (areas, cities, houses,...), Robotics, Information extraction, Social networks, Debugging.....

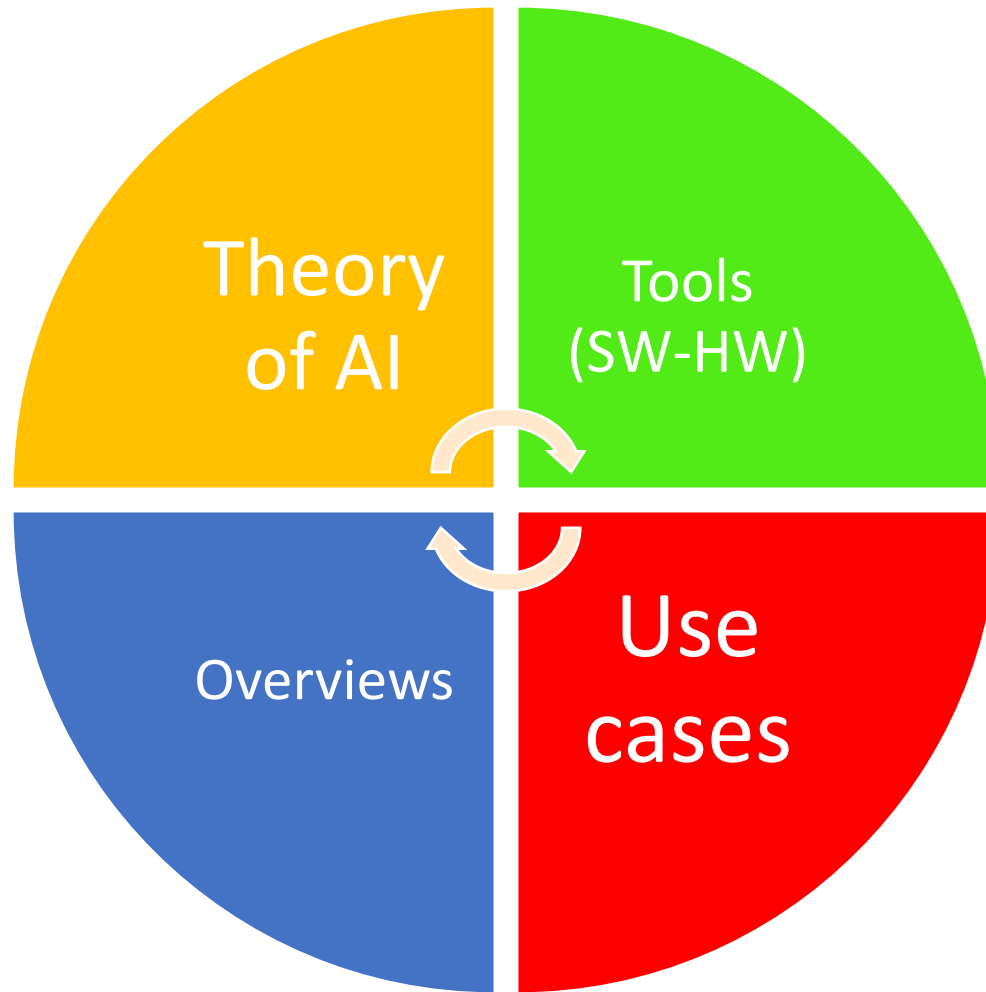
- **Your application!**

- Be smart and clever, AI can be applied in every task and activity of your project and application providing benefits



The course:

4 different type of contents

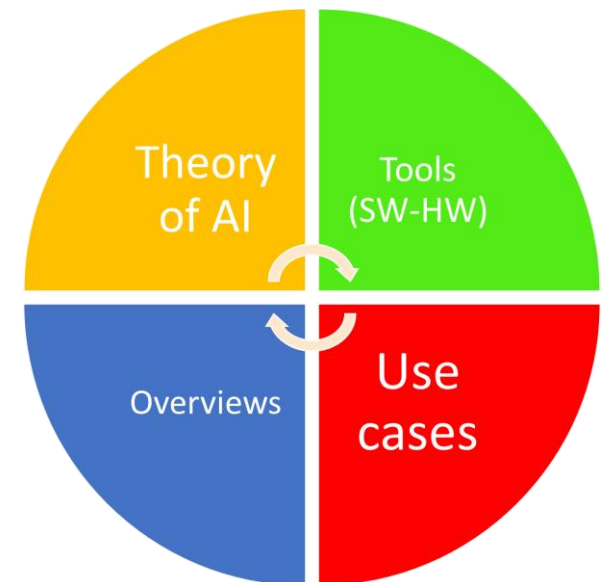


Some time
the difference
is «fuzzy»...

1. Theory of AI



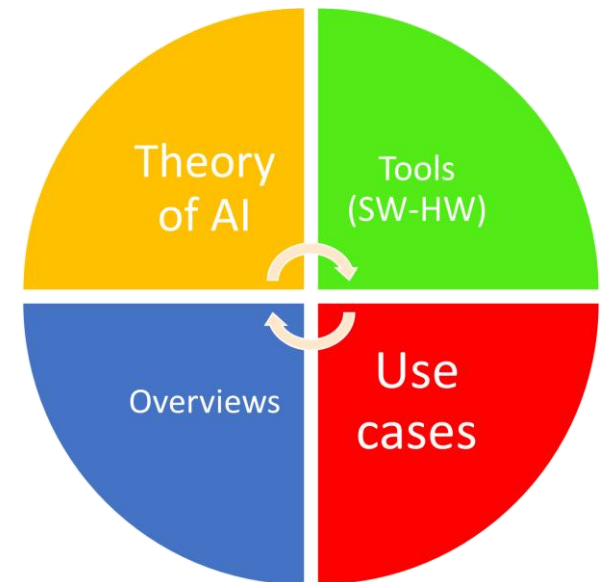
- Not limited to AI and neural networks
- Learn what we need to understand what's «under the hood»
- We will consider many topics
 - Pattern matching
 - Learning
 - Data analysis
 - Data processing
 - Feature extraction and fusion



2. Tools (SW-HW)



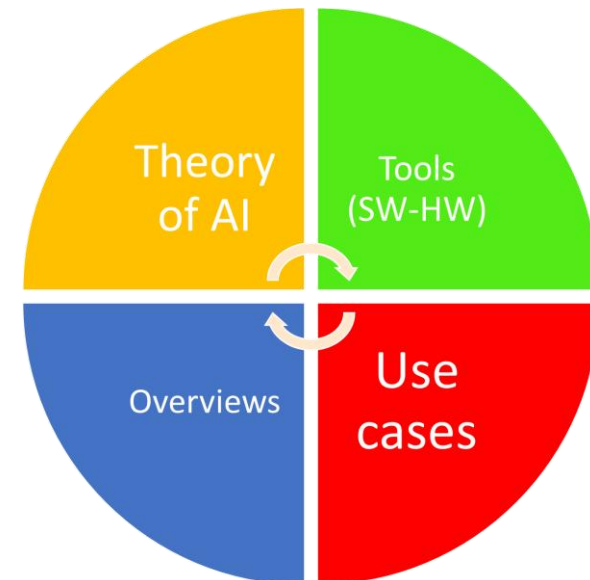
- What we need in our «toolbox»
- SW
 - Data analysis
 - Data processing
 - Data visualization
 - Learning environments
 - Flexibility, do not fall in love with one!
 - Matlab, Tensorflow, GoogleColab,...
- HW
 - CUDA and dedicated boards
 - Edge and fog computing



3. Overviews

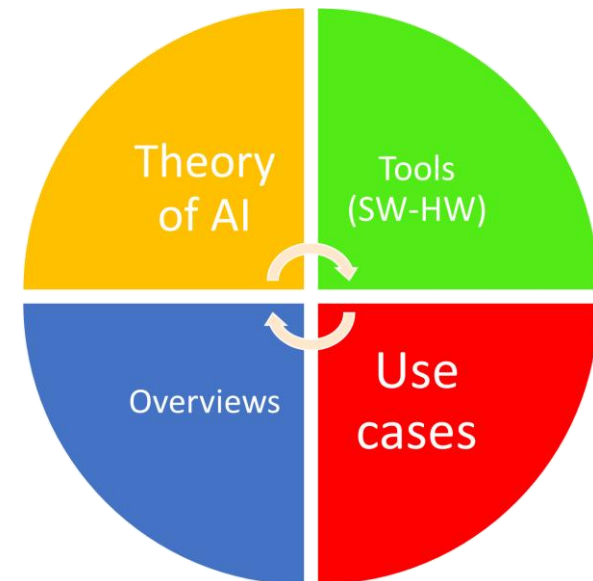
- Improving the knowledge about
 - State of the art
 - What can be done or not in different sectors and markets
 - What are the best tools
 - What kind of dataset we need
 - Limits of the techniques
 - General trends

Overview's information
are part of the course
and present in the exam



4. Use cases

- Examples
- Data management
- Inputs/Outputs
- Expected results
- Codes and snippets
- Usage of HW and SW tools



The course: different types of “work”

Class lessons

- Terms, Notions, Theory
- Design
- Examples and use cases
- Simulations of the exam

Laboratory (simulated)

- Simulations of inline problems and solutions
- Matlab, Tensorflow, Google Colab,...

Written exam

RE-DOWLOAD THE PDFs!

Lessons can be updated post-session to enhance readability, remove typos, add examples and links, etc.



Please remember to
reDOWNLOAD the PDFs at the
end of the course!

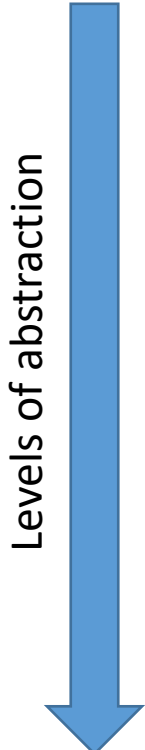




The IS4 course

Intelligent Systems for Industry, Supply chain and Environment
(in short “IS4”)

The structure of the course

- 
- Theory about AI models and learning methods
 - Classical Artificial Intelligence (pattern matching)
 - Recent developments (deep learning)
 - How to apply concepts to applications
 - Industrial
 - Environmental
 - Supply
 - Use cases

Be an expert in neural networks is enough! (?)

No

- Properties and differences between current models used in AI
- How to manage the data needed to train the models
- Avoid typical errors about the application of AI models
- Peculiarities about industrial applications in the different sectors
- Creating your own “toolbox”
- Deep learning models are just a part of the business

In brief...

Machine Learning = Deep Neural Networks

NO!!

If the only tool you have
is a hammer, **you tend to
see every problem as a
nail**



Abraham Maslow

Not just about training NNs

Some other skills needed in IS design

- State of the art
 - Provide the background historical (previous methods..)
 - Methodologies/Best practices
 - Analyze research trends
 - Describe what is present today in the literature and in the market
- Aggregate-fuse data
 - Can I add any knowledge, data, or feature to improve the learning?
- Comparisons (Fair)
 - How to compare methods and performances?



Not just training neural networks (2)



- Find features in common, find distinguishing features from similar/different domains/applications
- Create tables, charts, plots using the available data
- **ONLY AFTER THAT....**
Start train, validate and deploy models



EXAM

How to prepare yourself



Exam



- It will focus on the main topics discussed during the course with respect to

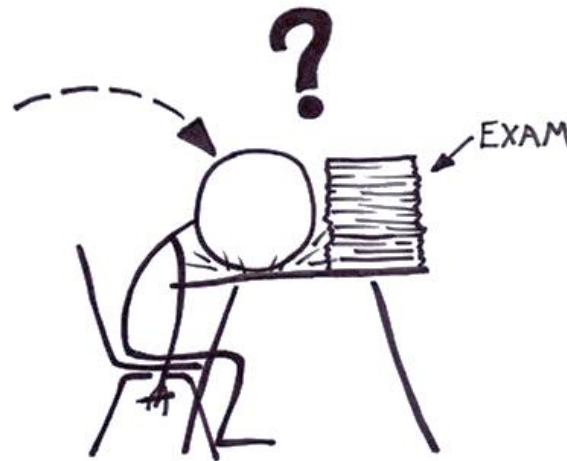
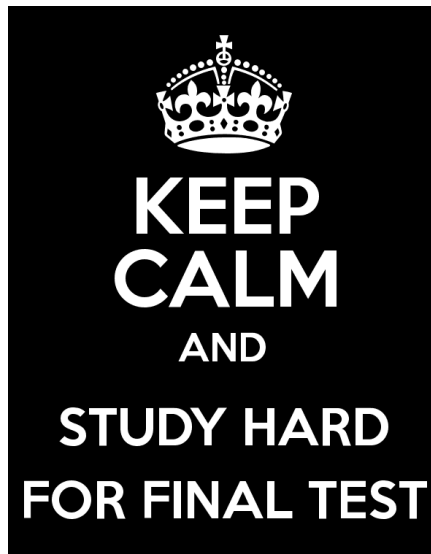
- Theory
- Overviews
- Use cases
- Tools
- Applications
- Design of simple cases



- **HINT:** browse all slides of the course and ask yourself: **what can be asked in this slide?**

The final exam

- **Written exam only (no projects)**
- **During the course will be given**
 - more details on how to face the exam
 - **simulations** of the exam



Exam structure

- The exam will be given in the laboratory in front of the PC
- It can be considered as a written exam
- No oral, No projects
- Topics of the test
 - Theory
 - Design cases
 - Code interpretation (as seen in class)
 - Use cases
- All questions/answers are based on the printed material of the course.



Exam structure

- It's a multiple-choice test
 - 1 single correct answer (weight = 1)
 - N wrong answers (weight = 0)
 - ~~1 (rarely 2) very bad answer (weight = -1)~~
No more negative weights
 - Final normalization of the test score to marks
- Duration: about 1h
- Can I do the test every exam date? YES
- Can I bring books/PC/material/etc? NO

Strategic approach: Rule #0

- You are not registered to the exam
→ no exam *****no exceptions*****
- You must be registered.
- Do it now! **Write on your calendar**

Handwritten "june 2022" above the calendar grid.

SUN	MON	TUE	WED	THU	FRI	SAT
			1	2	3	4
	5	6	UNIMIA		9	10
	12	13	14	IS4	16	17
	19	20	21	22	23	24
	26	27	28	29	30	



No automatic registration If you fail. Register for every call!

Strategic approach: Rule #1

- Is there a golden rule? YES, FIRST OF ALL
DO NOT LOOK FOR THE RIGHT ANSWER
EXCLUDE THE WRONG ONES

Ex. What is the best estimation of my age?

Searching for the right one	Excluding the wrong ones
A) 49 ← 67% confident B) 35 C) 12 D) 65	A) Don't care ← I choose this one B) 35 NO 75% confident C) 12 NO 99.9% confident D) 65 NO 80% confident

- Rule #1b: “Do you have doubts?”
→ choose the answer that “**looks less wrong**” to you

Strategic approach: Rule #2

- How can I improve my study of the material?
 - 1) Browse all slides of the course and ask yourself: what can be asked in this slide?
 - 2) BE SPECIFIC
 - 3) Do not use the simulations of the exam to study
 - 1) That is not the purpose!
 - 2) Use the simulation to understand
 - 1) how to face the questions and reason to find the solution
 - 2) If you are study profitably (see Strategic approach 1,2,3)

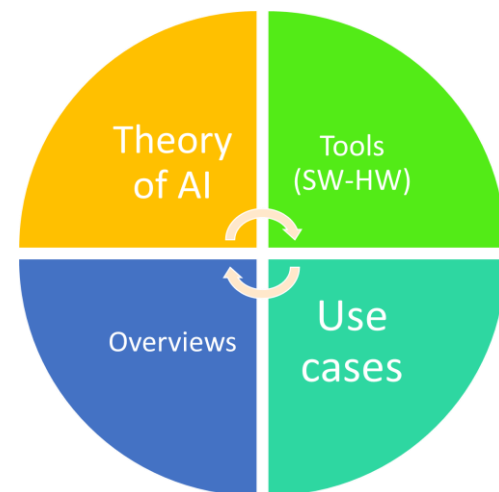


Overviews

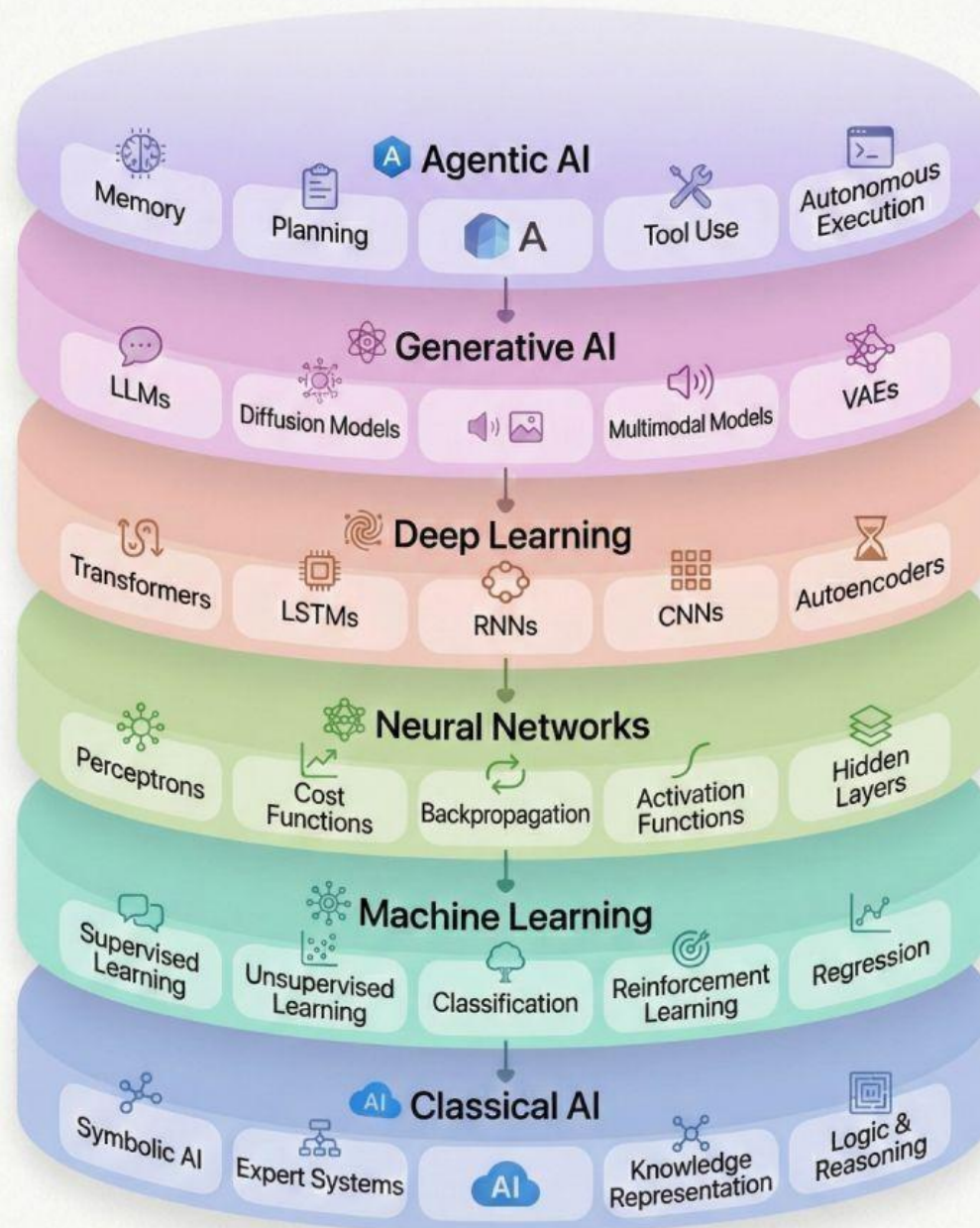


Major AI Advancements

An incredible acceleration!



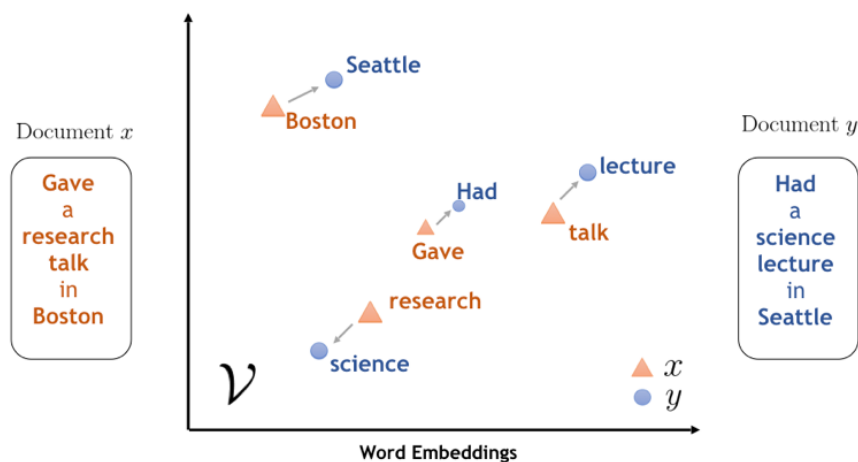
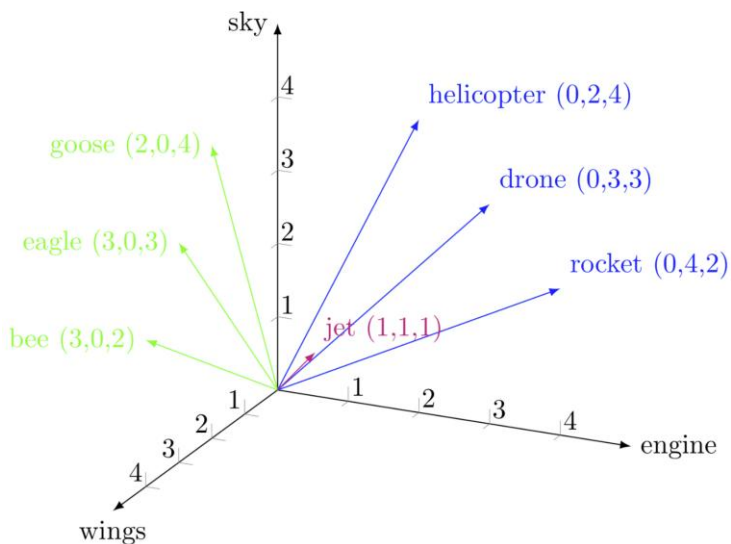
Layers of AI



Some recent improvements

(Word2Vec – 2013, Transformers e Contextual Embeddings - 2018+)

- So-called **word embeddings** that are often used as input for a neural network
 - A set of language modeling and feature learning techniques in natural language processing (NLP) where words or phrases from the vocabulary are mapped to **vectors** of **real** numbers



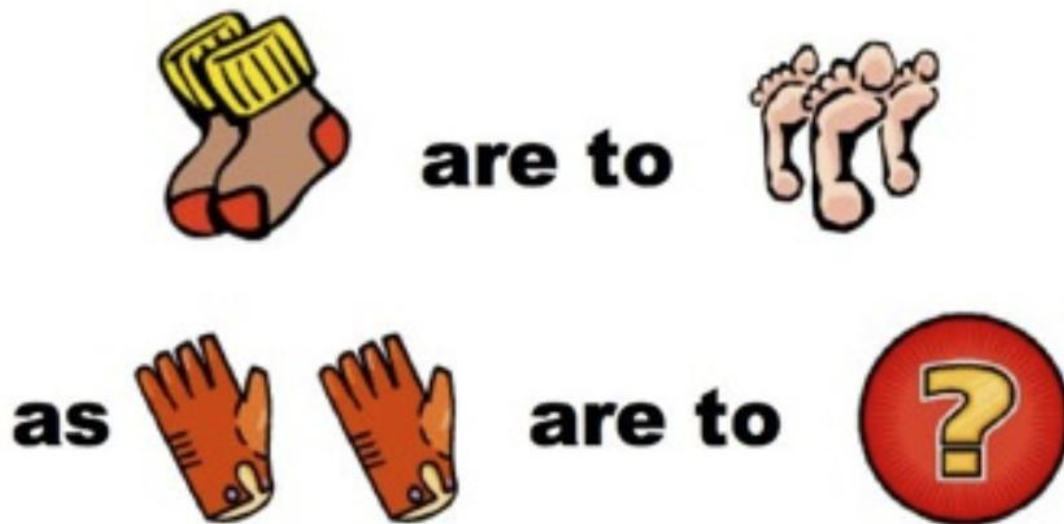
Some recent improvements (2)

- Solving **analogy puzzles**

(Frank Schilder, Armineh Nourbakhsh,
Thomson Reuters)

- “Paris is to France as Tokyo is to ?,”

-> producing the correct answer, “Japan.”



Examples

Also an state-of-the-art AI language model based on Generative Pre-trained Transformer like chatGPT can solve puzzles



Paris is to France as Tokyo is to



Japan.



cars are to city as ants are to



colony.



socks are to feet as gloves are to



hands.

Formicaio (Ant colony)

Other recent breakthrough

AI reaching International Mathematical Olympiad gold-level reasoning (2025)

- Models from OpenAI and Google DeepMind:
 - Achieved **gold medal level** at the **International Mathematical Olympiad**
 - Solved **5/6 problems** using **natural language reasoning**
- First time AI shows **human-level symbolic reasoning pipeline**

IMO human vs AI			
Entity	Year / setting	Score (out of 42)	Equivalent level
Typical human contestant	IMO 2024	16.0	Below bronze on that year's cutoffs
Typical human contestant	IMO 2023	17.7	Around bronze range in many years
Bronze medal human	IMO 2025 cutoff	19+	Bronze
Silver medal human	IMO 2025 cutoff	28+	Silver
Gold medal human	IMO 2025 cutoff	35+	Gold
Google DeepMind AlphaProof + AlphaGeometry 2	IMO 2024-style evaluation	28	Silver
Google DeepMind Gemini Deep Think	IMO 2025 official grading	35	Gold
OpenAI model	IMO 2025 non-official external grading	35	Gold-equivalent claim

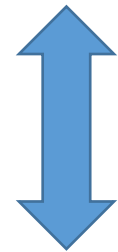


Artificial VS Natural brain

- Neural networks can already carry out complex cognitive tasks but...
- Current artificial networks are somewhere between the neural capacity of a bee's and a frog's brain.
 - The simple worm *C. elegans*, has 302 neurons and about 7000 synapses
- Human brains are at least **several orders of magnitude larger** than current neural networks
 - But the human brain has about 10^{11} neurons, and more than 10^3 synapses per neuron.... (Nature, 21 August 2019)
- Complex cognitive tasks are still far... but AIs are running fast and accelerating..

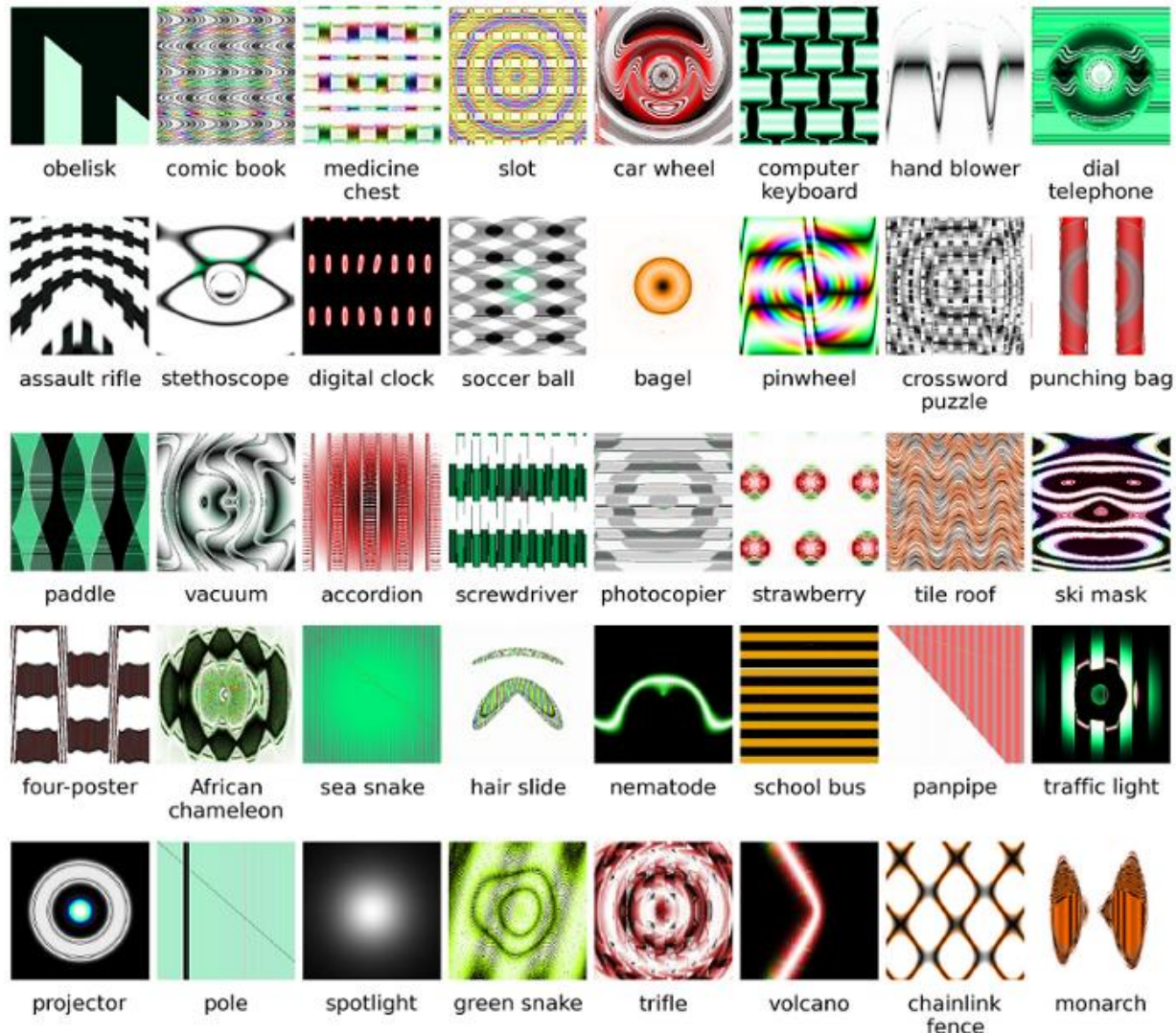


ML



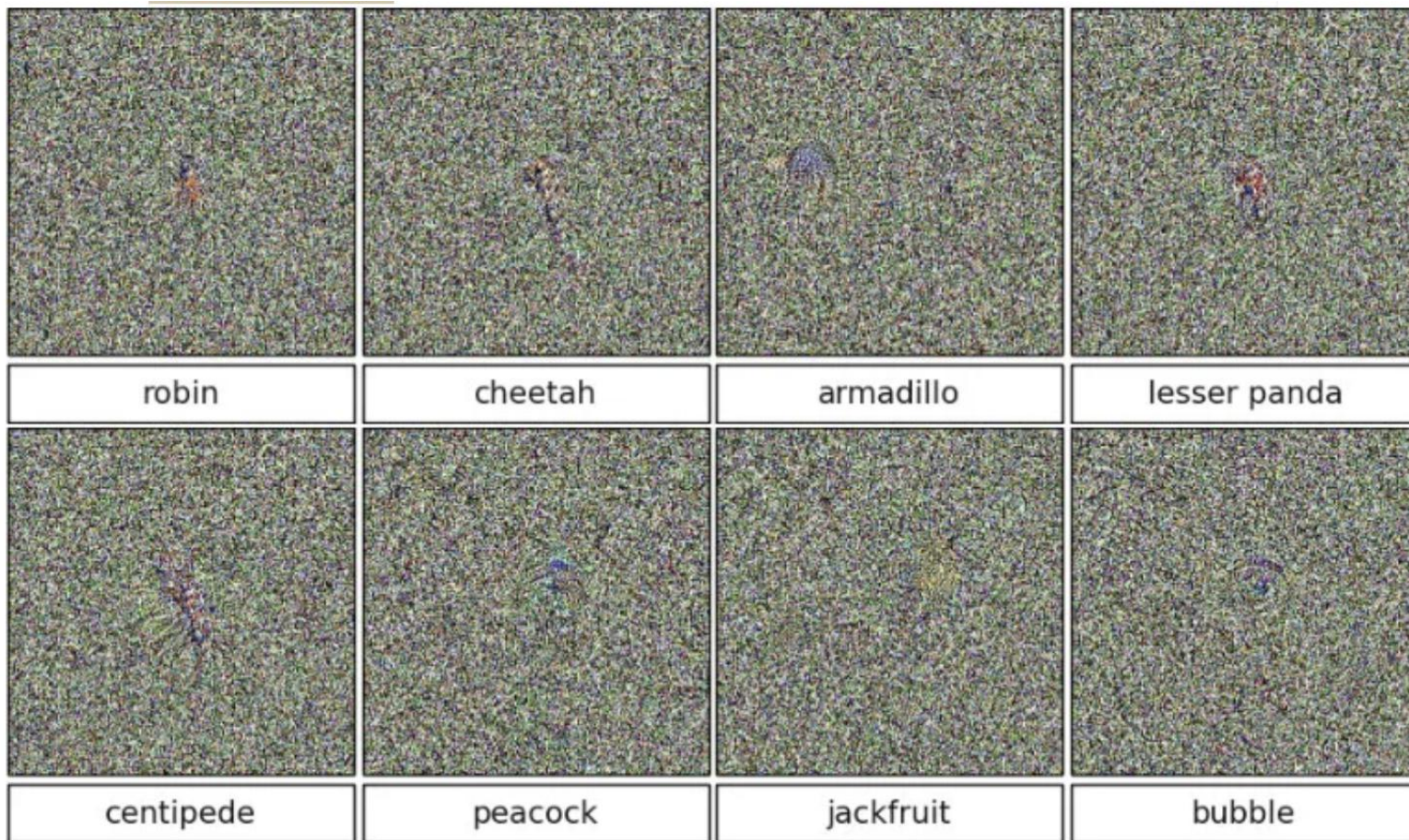
Not (yet) perfect

It is very easy to fool
a Deep Learning Neural Network
with artificially created images
(Anh Nguyen et al.)



AlexNet
"off the shelf"
neural network
trained on
ImageNet, a data
set of 1.3 million
images

Or even worst



AlexNet
"off the shelf"
neural network
trained on
ImageNet, a data
set of 1.3 million
images

Major AI Advancements in the Last recent years

- AI has made significant progress across various domains, revolutionizing industries and enabling new applications
- Starting to create profits..

Company	FY 2025 Revenue	FY 2025 Net Income	Key AI Growth Driver
Nvidia	\$130.5 Billion	\$60+ Billion (est.)	Data Center revenue grew 427% year-over-year.
Alphabet	\$402.8 Billion	\$132.2 Billion	Google Cloud revenue surged 48% due to AI demand.
Microsoft	\$281.7 Billion	\$101.8 Billion	Azure revenue grew 34% driven by AI platform trust.

The recent earnings underscore Big Tech's AI-powered momentum, with Microsoft, Meta, Amazon, Apple, Netflix and Google (Alphabet) leveraging advanced AI to drive robust year-over-year growth across ad targeting, personalisation, and scalable infrastructure.



Using models locally is a key feature for this course!

Major AI Advancements in the Last Years

Generative

Generative AI Models

- OpenAI: GPT-5.4 / GPT-5.4 Thinking
- Anthropic: Claude Opus 4.6
- Google: Gemini 3.1 Pro
- Meta: Muse Spark
- xAI: **Grok 4.1**
- In the last years, the introduction of GPT-4 by OpenAI, along with Google's Gemini and Meta's LLaMA models, has brought significant improvements in **text generation, reasoning, and multimodal capabilities**.
- These models are now widely used in **coding assistance, creative writing, and problem-solving across various industries**.

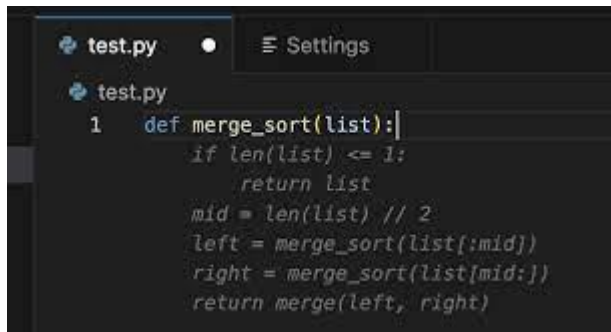


Major AI Advancements in the Last Years

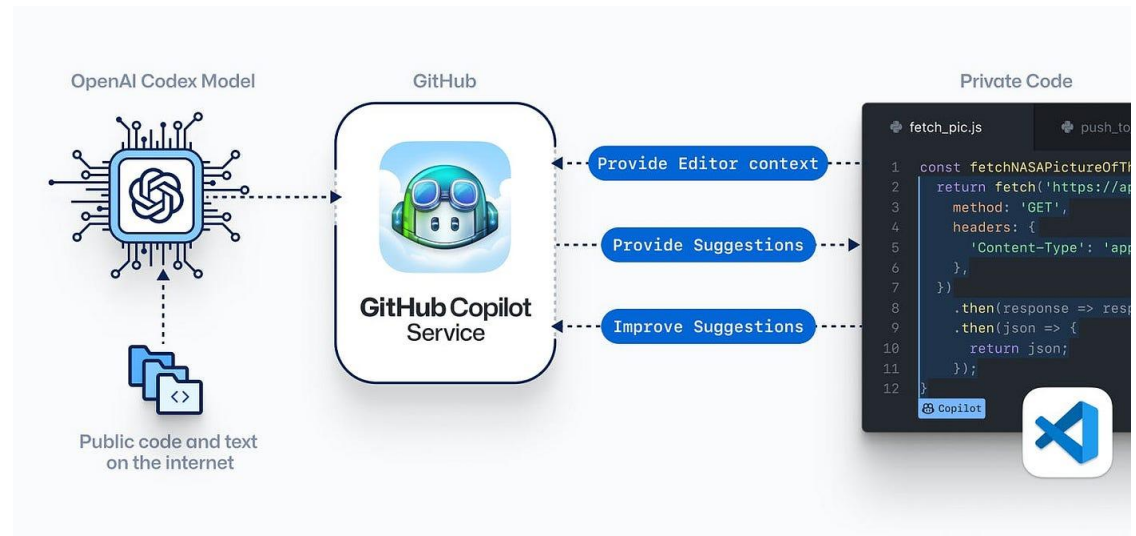
Coding

AI for Code Generation and Automation

- GitHub Copilot, Code Llama, and other AI-powered **coding assistants** have revolutionized software development, boosting productivity and reducing errors.
- AI-driven **debugging and optimization** tools are making software development more efficient and accessible.



```
test.py • Settings
test.py
1 def merge_sort(list):|
    if len(list) <= 1:
        return list
    mid = len(list) // 2
    left = merge_sort(list[:mid])
    right = merge_sort(list[mid:])
    return merge(left, right)
```

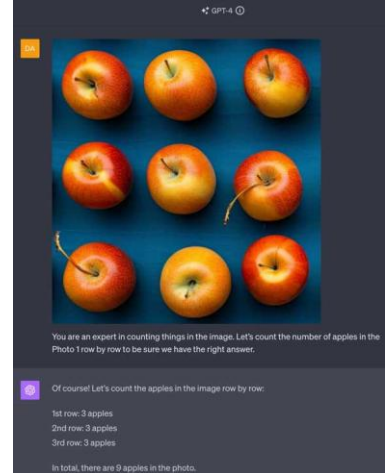


Major AI Advancements in the Last Years

Multimodal

Multimodal AI

- AI systems have evolved to handle multiple types of inputs, including text, images, audio, and video, simultaneously.
- Notable models include GPT-4V (vision support), Google Gemini's multimodal capabilities, and Meta's advancements in unified AI systems.

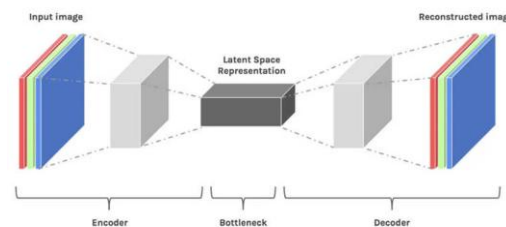


Major AI Advancements in the Last Years

Diffusion and Sci Discovery

Diffusion Models for Image & Video Generation

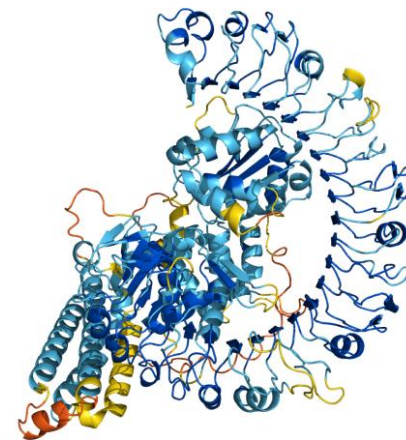
- Stable Diffusion, Midjourney, and OpenAI's DALL-E 3 have dramatically improved the quality and flexibility of **AI-generated images**.
- **AI-generated video** content is advancing with tools like Runway ML and Pika Labs, offering new possibilities for digital media creation.



A “diffused” N. Cage

AI for Scientific Discovery

- **AlphaFold 3** by DeepMind has further refined protein structure prediction, aiding drug discovery and biological research.
- AI is increasingly used in **materials science** for discovering new compounds and accelerating innovation in chemistry



Starting from 20 +2 amino acids... you can create all proteins

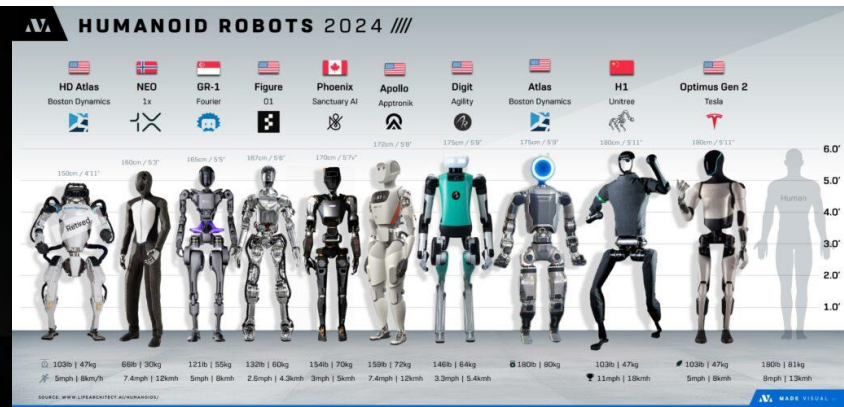
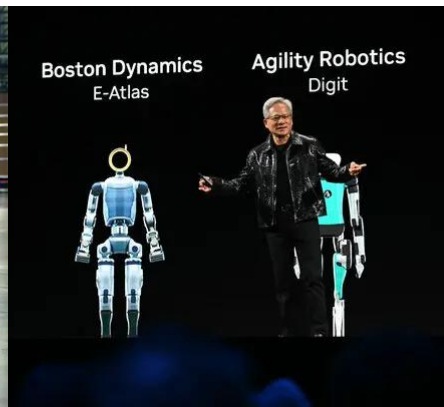


Major AI Advancements in the Last 2 Years

Robotics

Real-time AI in Robotics and Automation

- humanoid robot (e.g., Tesla's Optimus) continues to improve in dexterity and autonomy, moving towards practical applications in manufacturing.
- AI-powered robotic systems are enhancing industrial automation, logistics, and supply chain efficiency.

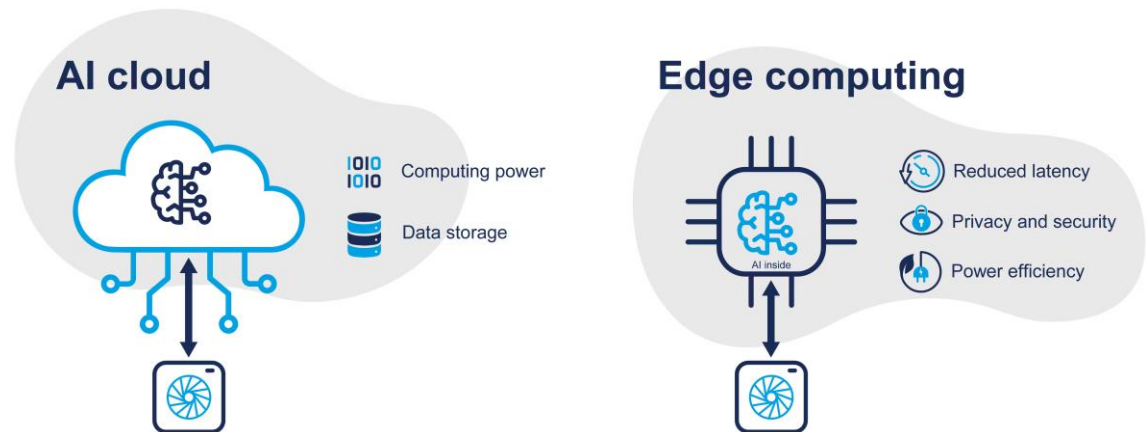


Major AI Advancements in the Last 2 Years

Edge AI

Edge AI and On-Device Processing

- Advancements in hardware allow AI models to run efficiently on mobile devices and edge computing platforms.
- Apple, Google, and Qualcomm have developed AI acceleration chips for smartphones, enabling real-time processing without cloud dependence.

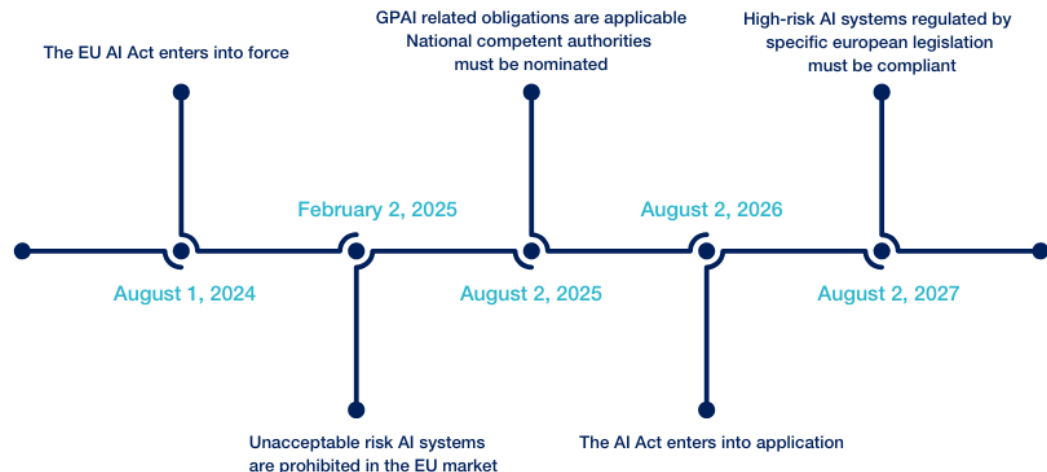
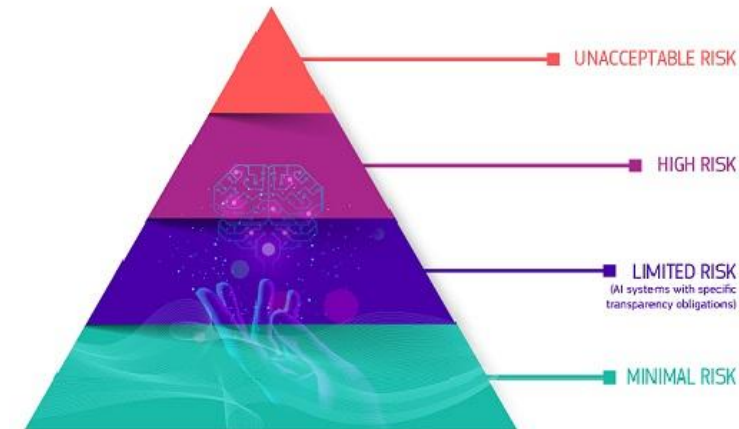


Major AI Advancements in the Last 2 Years

AI Act

AI Regulation and Ethical Considerations

- The European Union's AI Act has set a precedent for AI governance and regulation.
- Governments and organizations worldwide are actively working on responsible AI deployment and bias mitigation strategies.

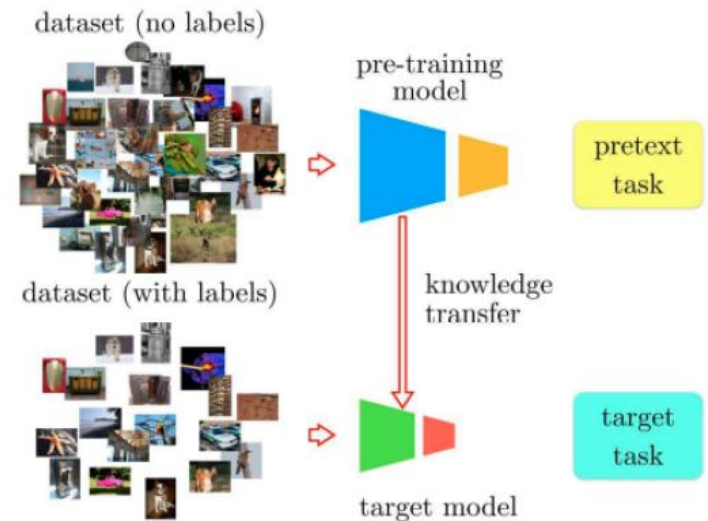


Major AI Advancements in the Last 2 Years

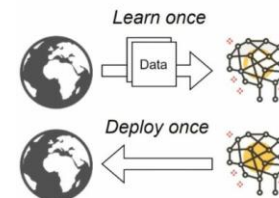
Self-Supervised and Continual

Self-Supervised and Continual Learning

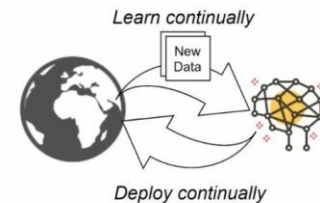
- AI models are increasingly able to learn from unlabelled data and adapt to new tasks with minimal human intervention.
- Self-supervised learning techniques are applied in medical imaging, such as AI models detecting anomalies in X-rays without requiring extensive labeled datasets.
- Speech recognition systems like Meta's wav2vec 2.0 have shown improved accuracy by leveraging self-supervised learning, reducing the need for human-labeled transcripts.
- AI in autonomous systems, including self-driving cars, is utilizing continual learning to adapt to new environments without requiring complete retraining.



Static ML



Adaptive ML



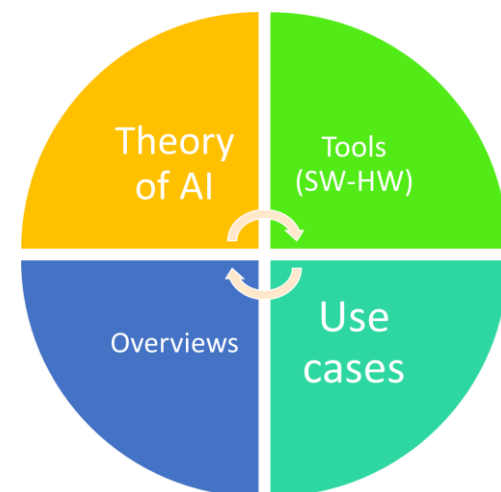


Tools (SW-HW)

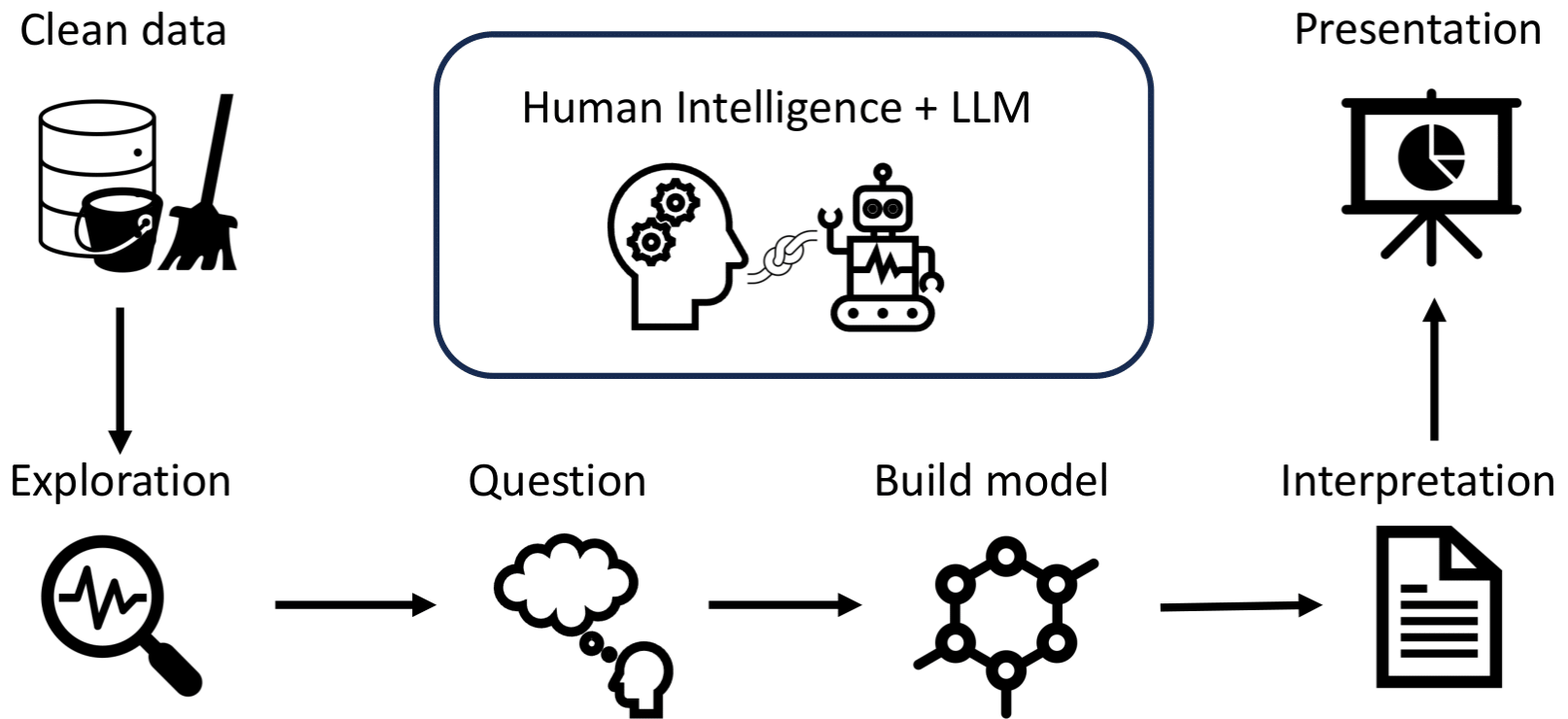
Can I use chatGTP?

You can benefit of LLM in the design of Intelligent Systems!

.....But with some warnings....



It's a good idea to use LLM to design IS (with some warnings)



Using LLMs as a Personal Designer for IS: Ideas and brainstorming

Useful Features for Intelligent System Design

- **Idea Gen. & Prototyping**
LLMs can help **brainstorm** and refine initial concepts for ISs.

To improve explainability in my models, it is better to use a neural network or a decision tree?

For improving explainability in your models, a **Decision Tree** is generally a better choice compared to a **Neural Network**.

Why Decision Trees Are More Explainable

1. **Interpretable Structure:** Decision trees have a hierarchical structure where each decision node represents a clear rule, making it easy to follow the reasoning behind a prediction.
2. **Feature Importance:** They provide direct insight into which features are most important for making decisions.
3. **White-box Model:** Unlike neural networks, decision trees are transparent—users can see the exact decision-making process.
4. **Rule Extraction:** The model can be converted into a set of IF-THEN rules, which makes it easy to communicate findings to non-technical stakeholders.

Challenges of Explainability in Neural Networks

1. **Black-box Nature:** Neural networks have complex, non-linear relationships between inputs and

Using LLMs as a Personal Designer for IS: Coding Assistance

Useful Features for Intelligent System Design

- **Code Assistance**
Capable of generating Python scripts for AI models used in industry, environment, and logistics.

in keras what is function call to create a convolutional neural network with 256x256 pixel input image and 25 3x3 kernel filter

In Keras, you can create a **Convolutional Neural Network (CNN)** with a 256x256 pixel input image and 25 filters of size 3x3 using the `Conv2D` layer. Here's the function call:

```
python

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, Input

# Define the model
model = Sequential([
    Input(shape=(256, 256, 3)), # Input shape for 256x256 RGB images
    Conv2D(filters=25, kernel_size=(3,3), activation='relu') # 25 filters of size 3x3
])

# Print model summary
model.summary()
```

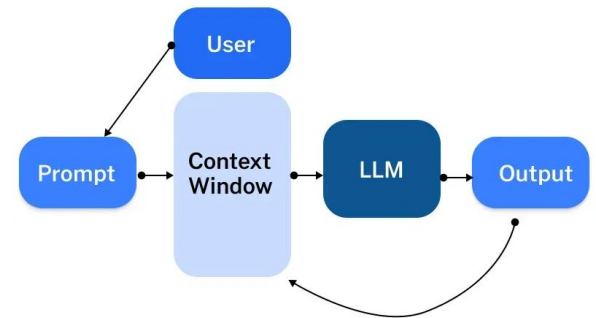
Explanation:

- `Input(shape=(256, 256, 3))` → Specifies the input shape of the images (256x256 pixels with 3 c ●

Using LLMs as a Personal Designer for IS: **Data analysis and explanations**

Useful Features for Intelligent System Design

- **Data Analysis & Insights**
Can process large datasets, provide exploratory data analysis, and generate reports.
- **Documentation & Explanation**
Useful for creating technical documentation and explaining AI methodologies.



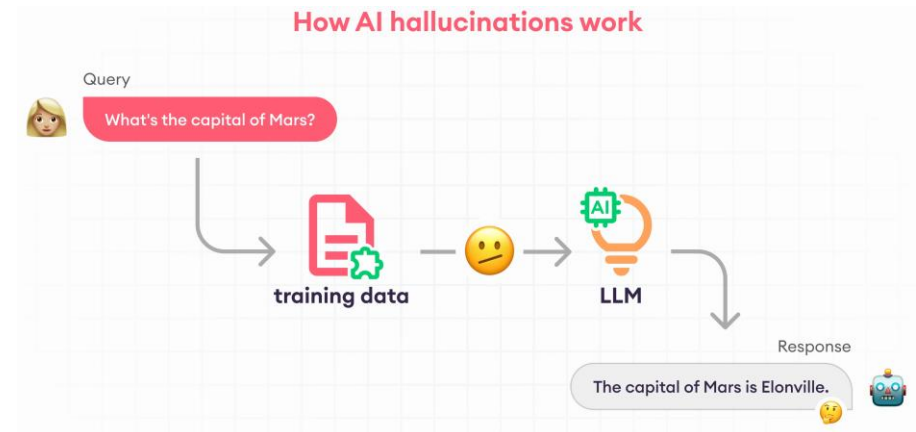
Using LLMs as a Personal Designer for IS: Hallucinations

Critical Considerations

- **Hallucinations**
LLMs can generate incorrect or misleading information, requiring careful validation.

"Provo ad ottimizzare la tabella che hai appena fatto, perché è inutile scrivere tutte le volte H, 2, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 30, 31, 32, 32, 33, 34, 34, 35, 36, 37, 38, 39, 40, 40, 41, 42, 41, 42, 43, 44, 44, 45, 46, 47, 48, 48, 49, 49, 50, 50, 51, 51, 52, 52, 52, 53, 52, 53, 53, 53, 54, 54, 55, 55, 56, 56, 57, 57, 57, 58, 58, 59, 59, 59, 60, 60, 60, 60, 60, 60, 60, 60,"

00:08



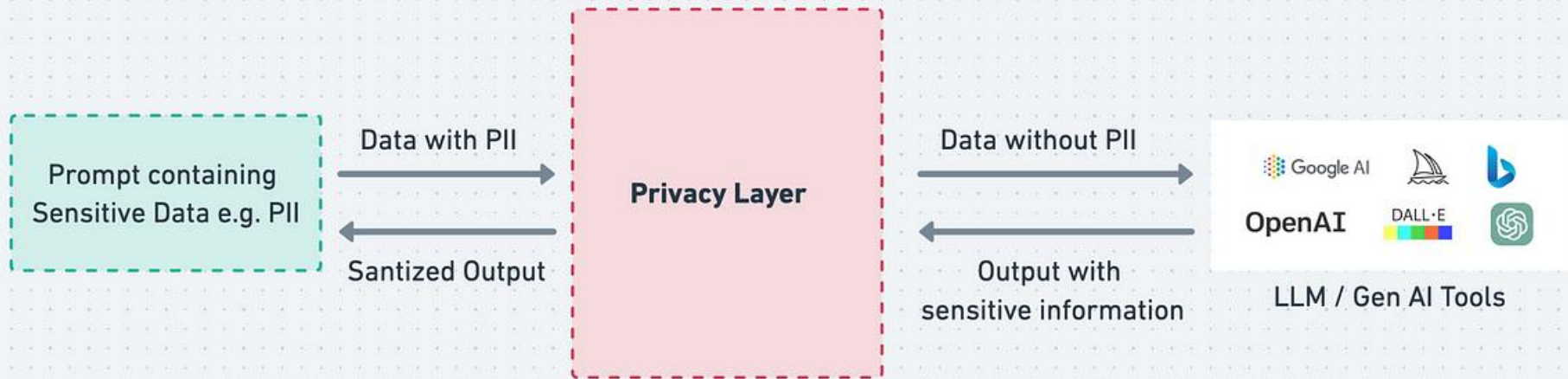
Hallucinations are the events in which ML models, particularly large language models (LLMs) like GPT-3 or GPT-4, produce outputs that are coherent and grammatically correct but factually incorrect or nonsensical.

Using LLMs as a Personal Designer for IS: **GDPR and Privacy**

Critical Considerations

- **Data Privacy & Security**

Using public LLMs for sensitive data may violate NDA or GDPR compliance.



Using LLMs as a Personal Designer for IS:

Biases

Critical Considerations

- **Bias in AI**
LLMs might reflect biases present in training data, requiring ethical oversight.
- **Dependency on Training Data**
Results may be outdated or not specialized for niche industrial applications.



Some basic testing of ChatGpt/LLMs

We will use version Chatgpt 3 to better understand “hallucination”



who is Fabio Scotti



Fabio Scotti is an Italian mathematician who is known for his contributions to the field of algebraic geometry. He has been a professor of mathematics at the University of Milano-Bicocca in Milan, Italy, since 2001.

Scotti's research focuses on the study of algebraic curves and surfaces, and their relation to other areas of mathematics such as topology and commutative algebra. He has made important contributions to the theory of moduli spaces of curves, and to the study of special classes of algebraic curves such as rational curves and plane curves.

Scotti has published numerous research papers and has given talks and lectures at conferences and universities around the world. He has also served as a referee for several mathematical journals and has been a member of editorial boards for several international mathematics publications.

In addition to his research and teaching, Scotti has also been involved in several outreach programs to promote mathematics education and research in Italy and beyond. He is considered as one of the leading mathematicians in Italy today.

what?! It's all wrong. There are no Fabio Scotti as described by ChatGPT3



Some basic testing of ChatGpt

Let's give it some **context**...



who is prof. Fabio Scotti at the university of Milan



Professor Fabio Scotti is a faculty member at the University of Milan in Italy. He is a Full Professor of Statistics at the Department of Economics, Management, and Quantitative Methods. His research interests include statistical inference, computational statistics, and data analysis.

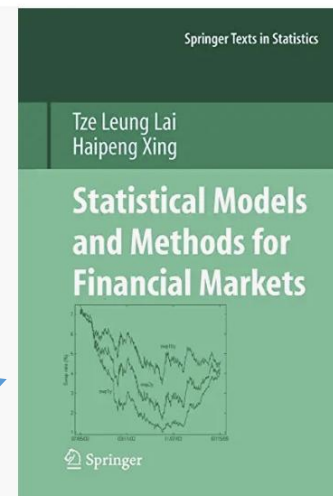
Professor Scotti has published numerous research articles in peer-reviewed journals, and he is also the co-author of the textbook "Statistical Models and Methods for Financial Markets." He has also received several awards and honors for his contributions to the field of statistics.

In addition to his academic work, Professor Scotti is also actively involved in consulting and collaborating with industry and government organizations.

?! Not present!!

RISULTATI

Scopri questi risultati.



Statistical Models and Methods for Financial Markets

Edizione Inglese
di Tze Leung Lai e Haipeng Xing

3.7 ★★★★★ ~ (10)

Copertina flessibile

78,88€

???

A matter of inputs/prompting...

During the course it will be clear that using the **proper inputs to train and validate a models is key** for the comprehension of the models' behavior and their effective deployment



what you can not do?



As an AI language model, there are certain limitations to what I can do. Here are some things I cannot do:

1. I cannot experience emotions or feelings.
2. I cannot have personal opinions or biases.
3. I cannot perform physical tasks or interact with the physical world.
4. I cannot understand sarcasm or humor in the same way humans do.
5. I cannot provide legal, medical, or financial advice.
6. I cannot violate ethical or moral standards or engage in illegal activities.
7. I cannot access information that is not publicly available or that is protected by privacy laws.

It's important to remember that while I can provide information and answer questions to the best of my abilities based on the data available to me, I am still a machine and may not always be able to provide the level of nuance or context that a human can.

"I am still a machine and may not always be able to provide the level of nuance or context that a human can."

We will discuss the input space of the model and the human nuances/context capability (Gestalt) in the next lesson

ChatGPT ver.4



who is prof. Fabio Scotti at the University of Milan



As of my knowledge cutoff date in September 2021, Fabio Scotti is an Associate Professor at the Department of Computer Science in the University of Milan, Italy. He has a background in computer engineering and specializes in the fields of computer vision, pattern recognition, and machine learning.

His research interests include:

- Biometric systems (e.g., fingerprint, face, and iris recognition)
- Image processing and analysis
- Artificial intelligence and machine learning applications
- Surveillance and security systems

Please note that the information provided may be outdated or incomplete. For the most current information about Prof. Fabio Scotti, it is recommended to visit his official university profile or personal website, or reach out to him directly through his professional email address.

Improved interpretability

reporting the limitations much better to the user



Lessons learned

Models are improving quite a lot and with a rapid speed

You must always be aware of the limitations by continuous testing and confrontation to the state of art!



You

who is fabio scotti at the University of Milan



ChatGPT

Fabio Scotti is a Full Professor at the Università degli Studi di Milano, Department of Computer Science. His areas of expertise include industrial, environmental, and biometrics informatics. Scotti has contributed significantly to his field, as evidenced by his extensive publication record and his role in leading the Industrial, Environmental, and Biometrics Informatics Laboratory at the university. His academic background includes a Ph.D. degree in computer engineering from the Politecnico di Milano, obtained in 2003. Since 2015, he has been an Associate Professor of computer science at the Università degli Studi di Milano [1].

Humor and serious concerns

AI is going to take over the world
AI:

Snail



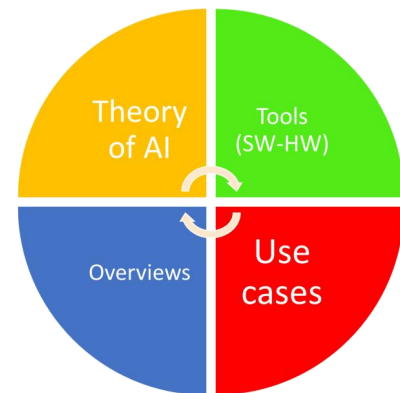
What if an autonomous vehicle with RGB cameras is not correctly processing this image?



THEORY

Natural interaction

One of the most relevant and limiting feature





Human-AI: Natural Interaction

Promises ...

- deep learning will permit to have a natural human machine interaction.
- Is now able (and will improve!) to **understand language, gestures, emotions, reason, interact** with humans in a more *natural (human-like) way*.

Human-AI: Natural Interaction

Key Modalities (for ISs)

Speech-Based Interaction and Natural Language Processing (NLP)

- Voice-activated healthcare diagnostics, smart home controls, vehicle navigation.

Gesture-Based Interaction

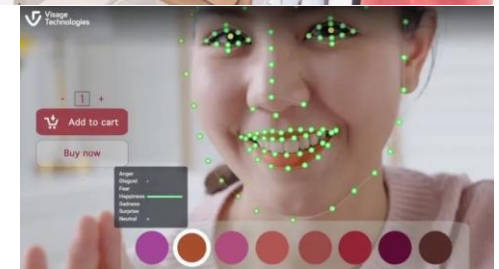
- Gesture-controlled industrial robots, AR training, sign language translation.

Haptic Interaction

- Surgical training simulations, wearable health devices, industrial diagnostics.

Visual & Emotional Recognition

- Customer sentiment analysis, student engagement tracking, mental health monitoring.



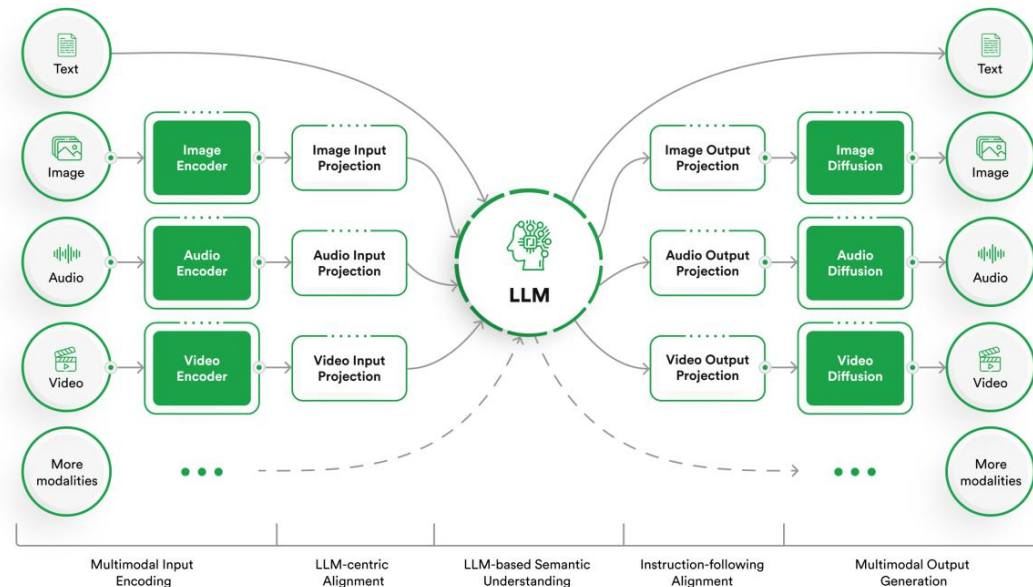
Human-AI: Natural Interaction Technologies (for ISs)

Context-Aware Sensing Adaptive Feedback Systems

- Predictive maintenance in factories, adaptive learning platforms, smart city traffic.



Multimodal Interaction (see previous slides)



Natural interactions limits

Discussion on NLP (speech)

Almost there improving

- Even though now has an **error rate <5%** compared to more than 10% just a few years ago,
 - It means that 40 words of an 800-word article would be wrongly recognized....
 - But more importantly:
→ A single critical misunderstanding can invalidate the whole task
- *Andrew Ng*, a machine learning world expert, has suggested that **naturally interacting** with a computer will not happen until we reach error rates of **<1%** (meaning in real applications)

Andrew Ng: VP & Chief Scientist of Baidu; Co-Chairman and Co-Founder of Coursera; Adjunct Professor at Stanford University.

Natural interaction: example

Voice Recognition For Order Picking in Logistics

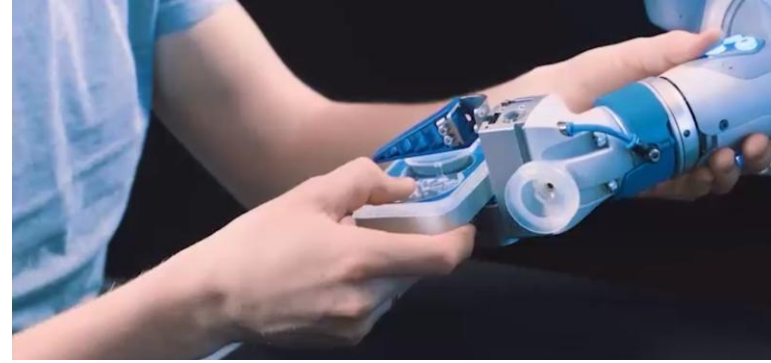


Voice picking systems for increased productivity, greater accuracy, and improved safety

<https://www.realtimelogistics.com.au/technologies/voice-picking-systems/>

Natural interaction

Advanced applications (not just voice...)



Cobots = Cooperative Robots

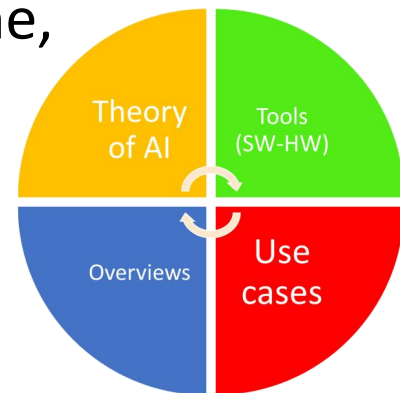




THEORY

Biases, improvements, limitations

“Similar to a child who grows and improves over time, the outcome depends on who is teaching.”

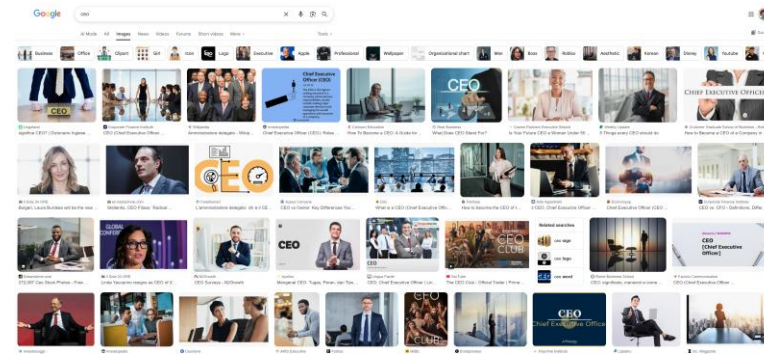


Some recent facts or improvements

- Neural networks are good at learning **human biases**
 - Cognitive biases are systematic patterns of deviation from norm or rationality in judgment

“You could mean bias in the sense of racial bias, gender bias. For example, you do a search for C.E.O. on Google Images, and up come 50 images of white males and one image of C.E.O. Barbie. That’s one aspect of bias.”

Daphne Koller



Harriet Lee-Merrion

Example of a different AI bias

- An algorithm is **latching onto something that is meaningless** and could potentially give you very poor results.
- Example
 - Predict fractures from X-ray images in data from multiple hospitals.
 - If you're not careful, the algorithm will learn to recognize which hospital generated the image.
 - Some X-ray machines have different characteristics in the image they produce than other machines, and some hospitals have a much larger percentage of fractures than others (e.g., large traumatology units).
 - And so, you could actually learn to predict fractures pretty well on the data set that you were given **simply by recognizing which hospital did the scan**, without actually ever looking at the bone.



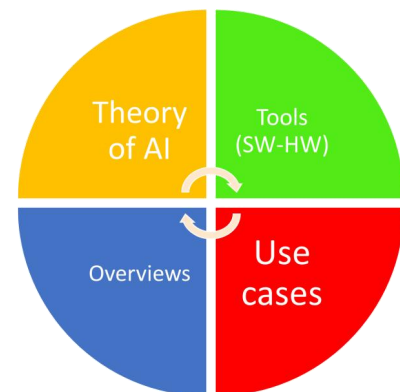
*Example from Daphne Koller
(Stanford and Coursera founder)*



THEORY

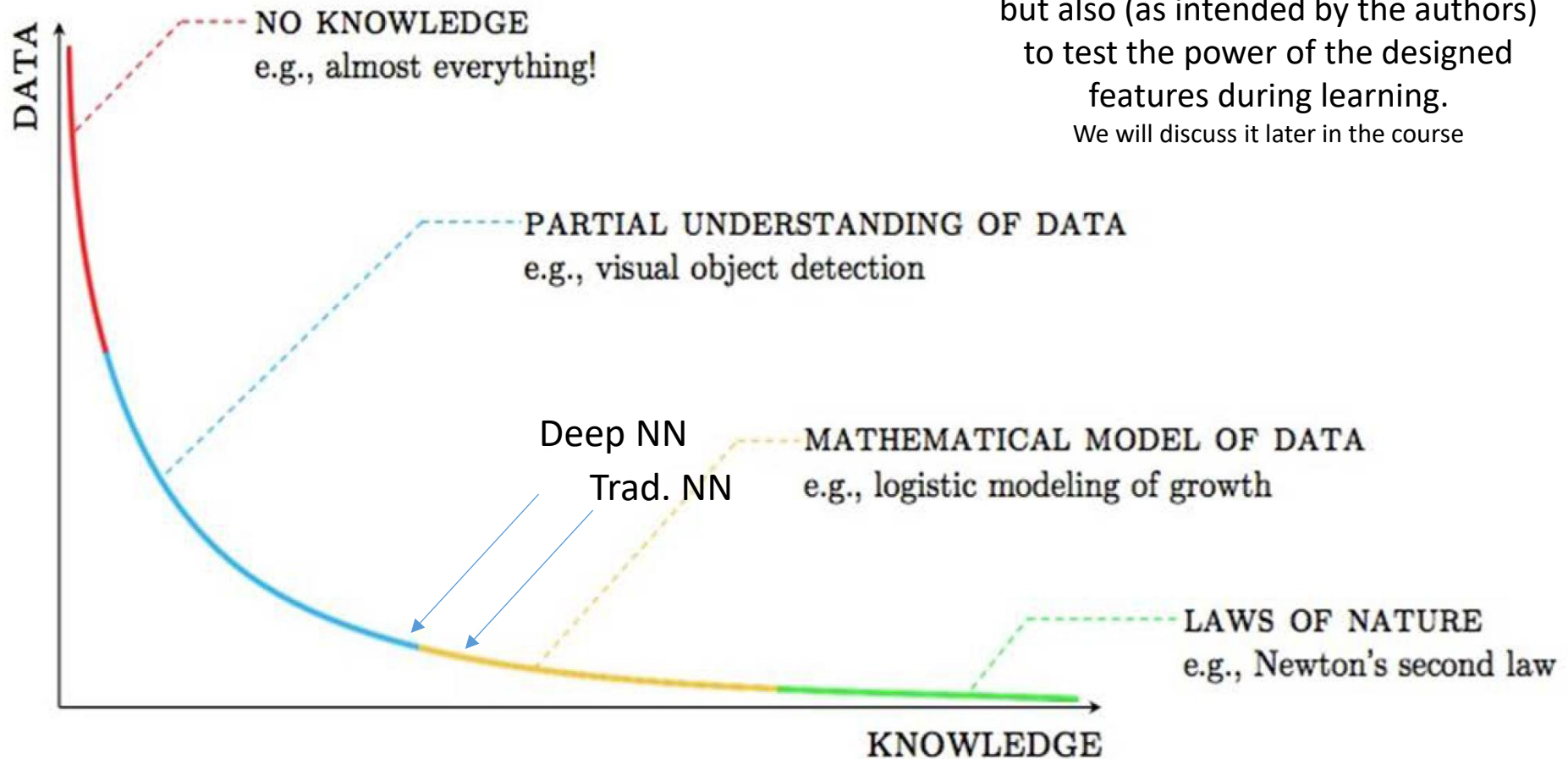
Data Knowledge Spectrum

Where is the knowledge in the data?



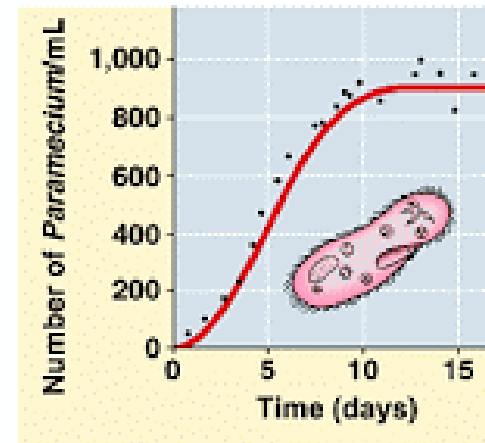
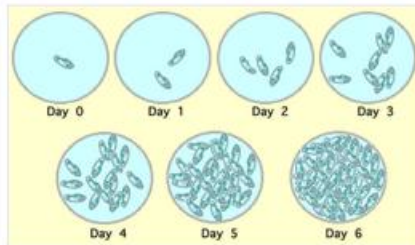
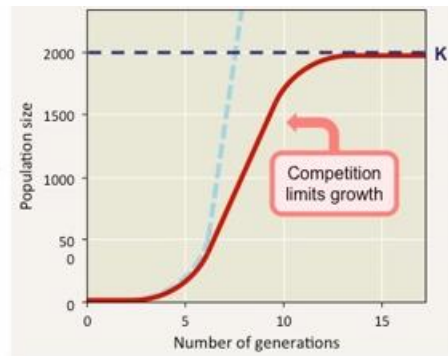
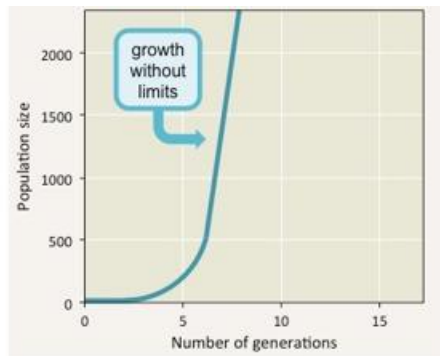
Data-Knowledge Spectrum

This plot can be considered as a general concept but also (as intended by the authors) to test the power of the designed features during learning.
We will discuss it later in the course



Data Knowledge Spectrum

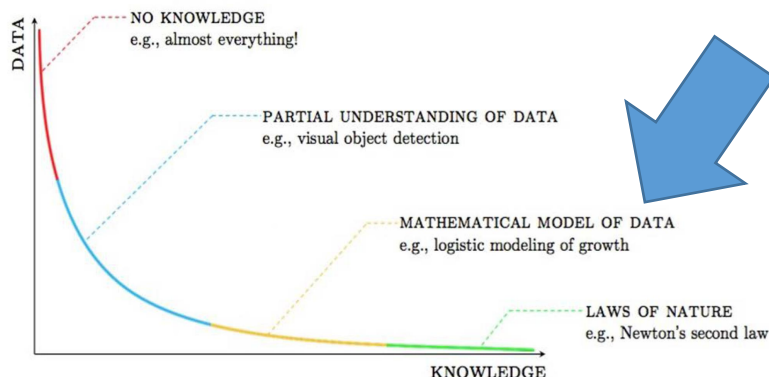
Example #1: Fitting simple data



(a) A *Paramecium* population in the lab

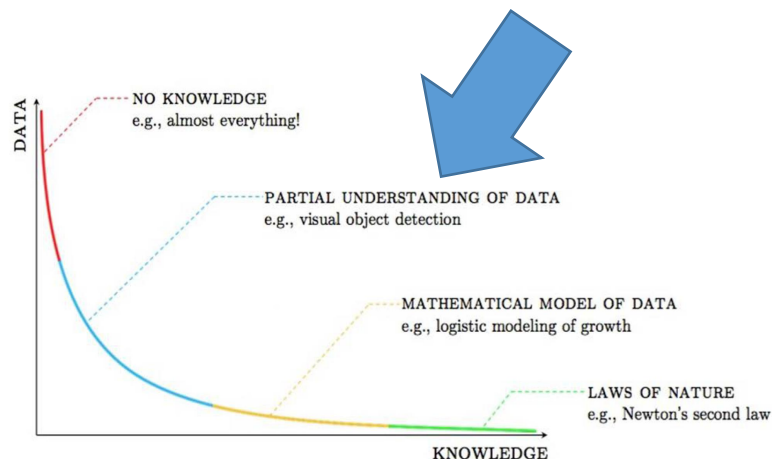
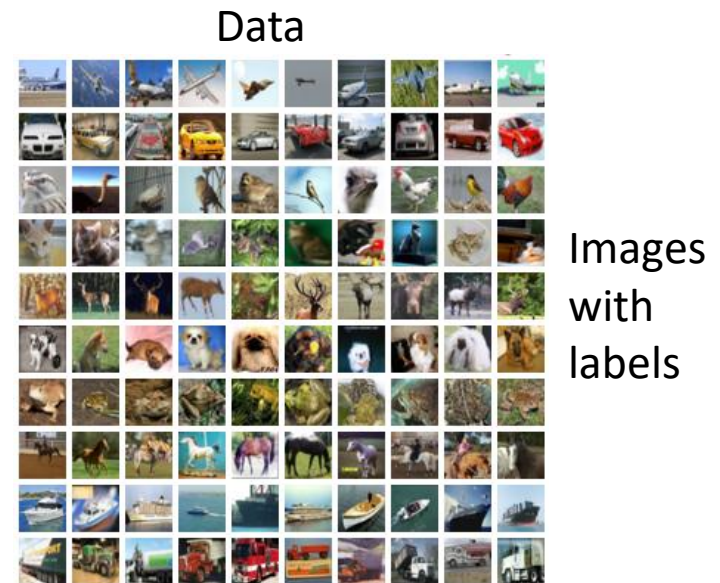
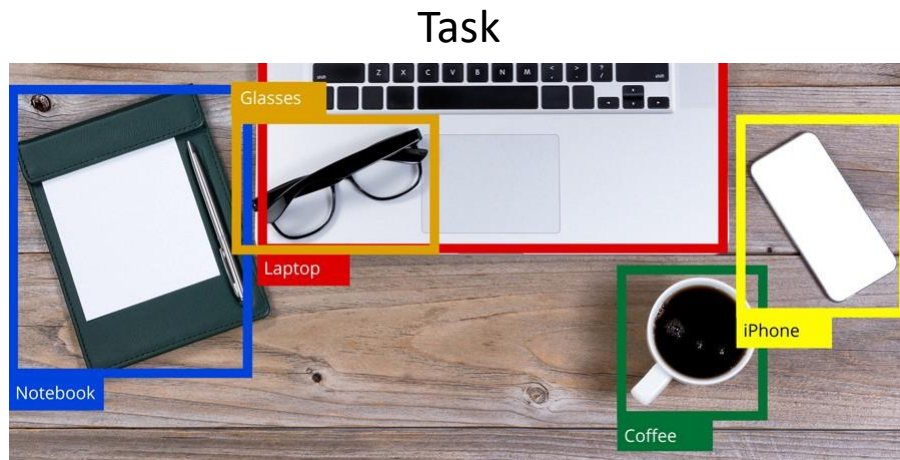
We have a very
specific and simple model
to be fitted with the data

It is not correct use a very large NN
to fit the data!



Data knowledge spectrum

Example #2: Visual object detection



We have not a specific model to be fitted with the data

We must use a large e general model to fit the data (at least we have data labels)
→ needs a lot of data to ne fitted correctly

Data knowledge spectrum

Example #3: Security attacks

Task

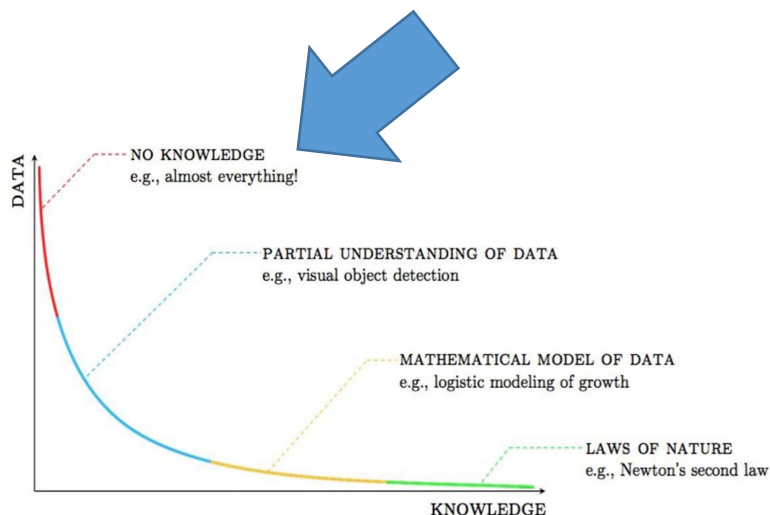
Data

What are the behaviors/features correct to identify attacks in this specific application?

(note: we have no labels,
so we can't create an attack classifier)

[illegible]

Million
of logs
with no
labels



We have not a specific model to be fitted
with the data with no labels

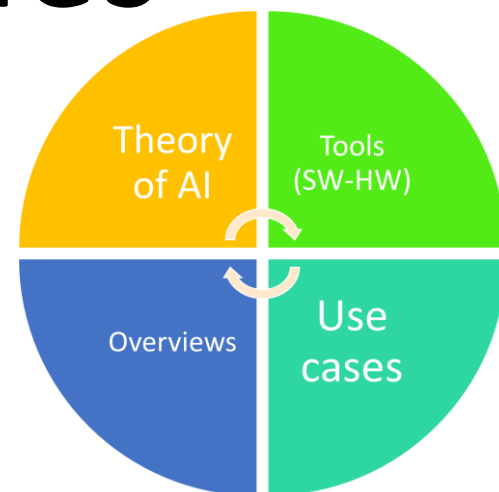
We must use a large & general models
capable to work in an unsupervised
fashion (no labels in data)



Use cases

Autonomous vehicles

A final example to summarize
some principal elements of the course



A comprehensive example: food delivery via autonomous vehicles

Ottobot from [Ottonomy](https://www.ottomomy.com) can maneuver through crowded places without GPS



Natural Language Processing and Umane Interaction

Image processing for Vision

Sensor processing
Self-calibration and diagnosis
AI Edge Processing

Crowd management

Navigation
capability and map
reconstruction



Many subsystems require AI, but some processing needs to be local (Edge)

Technologies and Models for Autonomous Systems

Navigation and Localization

- **Technologies:** SLAM (Simultaneous Localization and Mapping), GNSS (GPS, RTK), Lidar, Camera-based localization
- **AI Models:** Deep Reinforcement Learning for path planning, Kalman Filters for sensor fusion

Actuation and Control

- **Technologies:** Brushless motors, hydraulic actuators, PID/MPC controllers
- **AI Models:** Adaptive Control, Reinforcement Learning for dynamic stability

Path Planning

- **Technologies:** Search algorithms (A*), 3D maps, predictive neural networks
- **AI Models:** CNN+LSTM networks for obstacle anticipation

Human-Machine Interaction (HMI)

- **Technologies:** NLP (Natural Language Processing), facial/gesture recognition
- **AI Models:** Transformer models (ChatGPT, BERT), Vision-based sentiment analysis

Diagnostics and Predictive Maintenance

- **Technologies:** Sensor fusion (IMU, vibrations, temperature), edge computing
- **AI Models:** Predictive models on time-series data (RNN, LSTM)

Sensing and Environmental Perception

- **Technologies:** Lidar, Radar, Stereoscopic cameras, Ultrasonic sensors
- **AI Models:** Object Detection (YOLO, Faster R-CNN), PointNet for Lidar data

Edge AI and Real-time Processing

- **Technologies:** FPGA, TPU, Nvidia Jetson for local inference
- **AI Models:** Quantized neural networks for energy-efficient computation

The example of autonomous systems summarizes many of the key functionalities required for a complex intelligent system.

Throughout the course, we will explore many of these applications and models in detail.

Main points



- Major AI Advancements in the Last 2 Years
- Going from the theory to the application you need:
 - Knowledge about the theory of AI models and methods (strengths and weakness)
 - Applications experience, knowledge of the state of the art, methodology to deal with data and models
 - Good toolboxes
- AI is greatly improving but strong limitations are present
 - Biases
 - Natural Interaction
 - Not explainable
- The less we know about the process, more data is needed to develop automatically the neural model
 - Data Knowledge Spectrum
- Course structure, Exam structure and golden rules